

PROBLEM STATEMENT 2 — SOLUTION DOCUMENT

ECONOMIC TIMES presents

AI-Driven Energy Supply Chain Resilience for Import-Dependent Economies

Supply Chain Intelligence · Energy Security · Geopolitical Risk

Prepared by

Adrish Kumar Das

+91 86173-06099

Official mail: adrishkumardas05.24@kgpian.iitkgp.ac.in

Personal mail ID: adrishkumardas2006@gmail.com

Indian Institute of Technology , Kharagpur

In the demo video I will go through the thinking journey , and made a prototype of that. So here is my final website : [Indra-Global-Risk-Measurement](#)

1. Executive Summary

India sources roughly 88–90% of its crude oil through imports, and 40–45% of that volume transits the Strait of Hormuz — a corridor currently under sustained geopolitical strain following the 2025 US–Iran standoff, renewed sanctions on Iranian oil exports, and repeated Red Sea shipping incidents. With a 4–6 week import lead time and national strategic reserves covering only a matter of days, a route committed to today can turn unsafe before the cargo ever arrives.

INDRA is a checkpoint-level risk screening system built on GDELT 2.0 event data. It scores every checkpoint along a candidate import route on a 0–100 Global Risk scale, flags any checkpoint at or above a 55-point threshold, and lets a route be compared against alternatives before a shipment is committed. Two lightweight, browser-based tools operationalize the model: a batch Risk Console for scoring a full GDELT export at once, and a Location Risk Lookup for querying the risk history of a single checkpoint by latitude and longitude.

This document sets out the problem context, the scoring methodology, the tools built, how the solution maps to the suggested building blocks in the problem statement, and the forecasting work still ahead.

2. Problem Statement

2.1 Context

India sources approximately 88% of its crude oil through imports, with 40–45% of that volume transiting the Strait of Hormuz. This is not a theoretical risk — it is a structural vulnerability that geopolitical events regularly stress-test. The 2025 US–Iran standoff sent Brent crude up over 8% within a single session, and renewed sanctions on Iranian oil exports, maritime security incidents in the Persian Gulf, and escalating Houthi attacks on Red Sea shipping lanes have kept the supply threat live.

India's Strategic Petroleum Reserves provide roughly 9.5 days of national consumption cover — a buffer that would be exhausted quickly in any sustained disruption. A McKinsey analysis of past energy supply shocks found that economies without automated rerouting and demand-management capability took an average of 47 days to stabilise, compared with those that had an integrated response intelligence layer built in.

Traditional supply chain planning tools were designed for predictable environments. They have little ability to model geopolitical scenarios in real time, dynamically evaluate alternative procurement corridors, or orchestrate a coordinated response across refiners, logistics providers, and strategic reserves. That intelligence layer needs to be built.

2.2 Challenge Statement

Design an AI-powered Energy Supply Chain Resilience system that monitors geopolitical and logistics risk signals continuously, models disruption scenarios, and generates executable procurement and rerouting recommendations — turning a reactive crisis response into a managed, anticipatory process.

2.3 Suggested Building Blocks

The problem statement outlines five illustrative components a full solution might explore:

- **Geopolitical Risk Intelligence Agent** — a multi-source agent ingesting news feeds, shipping AIS data, sanctions signals, and commodity price signals to score disruption probability by corridor and supplier, updated continuously.
- **Disruption Scenario Modeller** — simulates specific events (e.g. a Hormuz partial closure, an OPEC+ emergency cut, Red Sea suspension) and cascades the impact on refinery run rates, fuel prices, power sector stress, and GDP trajectory.
- **Adaptive Procurement Orchestrator** — an agentic system that identifies and ranks alternative crude sources and logistics routes, factoring in spot market pricing, tanker availability, port congestion, and refinery grade compatibility.
- **Strategic Reserve Optimisation Agent** — models optimal SPR drawdown schedules against supply-gap forecasts, refinery demand curves, and replenishment windows.
- **Supply Chain Digital Twin** — a geospatial simulation of India's full energy supply network, from wellhead to refinery to distribution, enabling continual what-if analysis.

3. Proposed Solution — INDRA Corridor Risk Engine

3.1 Solution Philosophy

Rather than attempting the full five-agent system in one pass, INDRA focuses on the layer every other component depends on: a reliable, quantified read of geopolitical risk at the specific checkpoints a shipment must pass through. The approach is deliberately narrow and operational — given a candidate route and its checkpoints, tell a planner today whether each checkpoint is currently safe to route through, using a transparent and inspectable scoring model rather than a black box.

3.2 Data Foundation

The engine runs on GDELT 2.0 event exports, using the following fields for every event: GoldsteinScale (a -10 to +10 cooperation/conflict intensity score), AvgTone (average media sentiment), NumMentions, NumSources and NumArticles (coverage volume), QuadClass (CAMEO conflict/cooperation classification), EventRootCode (CAMEO root category), and each event's geocoded latitude, longitude, and date. A GDELT export is cleaned and feature-engineered before scoring — this project's working pipeline processed roughly 106,000 raw GDELT rows into a structured, feature-complete dataset.

3.3 Risk Scoring Methodology (All the formulas I use in The Projects)

Each event is scored through a chain of intermediate metrics that build up to a single 0–100 Global Risk value. All normalization is min–max scaling within the batch being scored.

Metric	Formula (summary)	Purpose
GoldStienRisk	Normalize (max (0, -GoldsteinScale))	Inverts the GDELT Goldstein cooperation/conflict score into a 0–100 risk value.
ToneRisk	Normalize (max (0, -AvgTone))	Same inversion applied to average media tone.
Importance	Normalize (0.5·NumMentions + 0.3·NumSources + 0.2·NumArticles)	Composite of how widely an event was covered.

Conflict / MilitaryActivity	100 if QuadClass=4, 50 if QuadClass=3, else 0	Encodes CAMEO quad-class severity (material vs. verbal conflict).
Conflict_Score	$0.4 \cdot \text{Conflict} + 0.3 \cdot \text{GoldStienRisk} + 0.2 \cdot \text{ToneRisk} + 0.1 \cdot \text{Importance}$	Blended conflict-intensity indicator.
RootCodeSeverity	lookup table, CAMEO EventRootCode 1–20 → weight 5–100	Severity weight per CAMEO root category.
Severity_Score	$0.4 \cdot \text{RootCodeSeverity} + 0.3 \cdot \text{GoldStienRisk} + 0.2 \cdot \text{ToneRisk} + 0.1 \cdot \text{NumMentions}(\text{norm})$	Base severity of the event.
Adjusted_Severity_Score	$\text{Severity_Score} \times (1 + \text{ToneRisk} / 10)$	Amplifies severity when coverage tone is sharply negative.
SupplyChainScore	$0.35 \cdot \text{AdjSeverity} + 0.25 \cdot \text{GoldStienRisk} + 0.2 \cdot \text{MediaSentimentRisk} + 0.1 \cdot \text{NumSources} + 0.10 \cdot \text{ToneRisk}$	Logistics-relevant risk exposure.
PoliticalScore	$0.40 \cdot \text{AdjSeverity} + 0.25 \cdot \text{ToneRisk} + 0.20 \cdot \text{GoldStienRisk} + 0.15 \cdot \text{NumMentions}(\text{norm})$	Political-dimension risk exposure.
SanctionScore	AdjSeverity if EventRootCode ∈ {10,12,13,16,17}, else 0	Flags sanction / coercion / reject-class CAMEO events.
Global Risk	$\text{normalize}(0.3 \cdot \text{Conflict_Score} + 0.2 \cdot \text{GoldStienRisk} + 0.15 \cdot \text{MediaSentimentRisk} + 0.10 \cdot \text{MilitaryActivity} + 0.10 \cdot \text{SupplyChainScore} + 0.05 \cdot \text{PoliticalScore} + 0.10 \cdot \text{SanctionScore})$	Final 0–100 composite score per event, per checkpoint.

Table 1 — Global Risk scoring chain, applied to every event in a GDELT export.

3.4 Checkpoint & Radius-Based Screening

A planner supplies a checkpoint's latitude and longitude — a refinery, port, or a chokepoint such as the Strait of Hormuz — plus a search radius (or an exact-coordinate match for tightly geocoded facilities). The engine returns every GDELT event within that area: its date and Global Risk score individually, along with the checkpoint's peak risk and average risk across the full event history loaded.

3.5 Decision Threshold

A checkpoint is treated as cleared for routing when its Global Risk stays below 55. At or above 55, the checkpoint is flagged and the planner is expected to evaluate an alternate route rather than commit a shipment through it. This threshold is a starting heuristic, set deliberately in the middle of the scoring range, and is intended to be recalibrated against real freight, insurance, or delay outcome data as it becomes available.

Map of Middle East to India crude oil shipping routes, showing checkpoints at export terminals, chokepoints, and Indian import ports connected by red route lines



Figure 1 — Checkpoint screening logic applied per candidate route.

3.6 Route Selection Framework

Five candidate maritime routes are currently tracked between Gulf departure points and Indian ports. Three of the five pass through the Strait of Hormuz checkpoint (26.566° N, 56.250° E), making it the single highest-priority point to monitor across the route set.

Route	Description	Passes Hormuz Checkpoint?	Screening Priority
Route 1	Persian Gulf → Strait of Hormuz → Arabian Sea → West coast India	Yes	High
Route 2	Persian Gulf → Strait of Hormuz → Gulf of Oman → West coast India	Yes	High
Route 3	Persian Gulf → Strait of Hormuz → Southern Arabian Sea route	Yes	High
Route 4	Alternate Gulf departure avoiding Hormuz corridor	No	Standard
Route 5	Alternate Gulf departure avoiding Hormuz corridor	No	Standard

Table 2 — Candidate route set and Hormuz exposure.

For a given shipment, every checkpoint on every candidate route is scored. A route is preferred when all of its checkpoints clear the 55-point threshold; if a route's checkpoint is flagged, the planner moves to the next candidate and repeats the check, continuing until a route with an acceptable risk profile is found.

4. System Architecture & Tools Built

Both tools are single-file, browser-based applications — a cleaned GDELT CSV or XLSX export is loaded directly in the browser, scored client-side in JavaScript, and never leaves the user's machine. There is no server component and no external dependency beyond the charting and spreadsheet-parsing libraries loaded from a CDN.

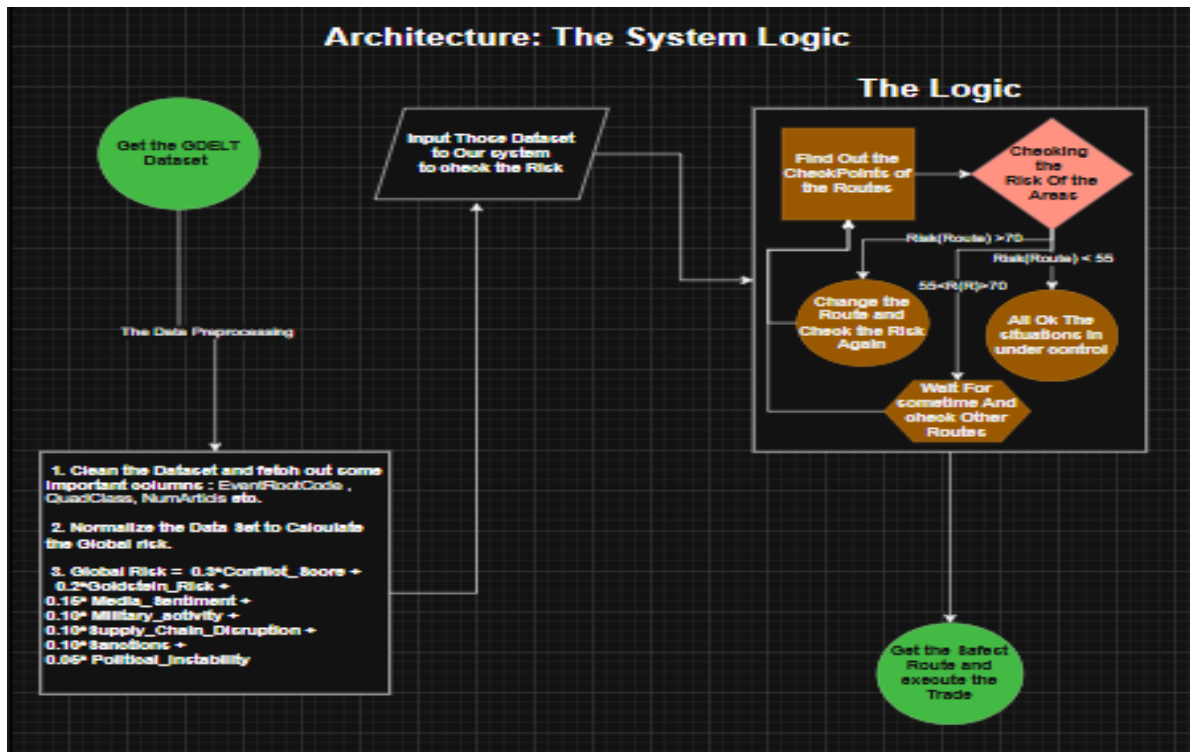


Figure 2 — Data flow from raw GDELT export to a routing decision.

4.1 INDRA Risk Engine Console

Accepts a full GDELT export and scores every row at once. Surfaces a risk distribution histogram, an average-risk breakdown by CAMEO QuadClass, a ranked table of the ten highest-risk events in the feed, and a searchable, sortable, paginated view of the full scored dataset with CSV export.

4.2 INDRA Location Risk Lookup

Purpose-built for the checkpoint workflow in Section 3.4 and 3.5. A planner loads the same scored feed, maps the date and latitude/longitude columns (auto-detected against common GDELT naming conventions, editable if wrong), and queries a specific checkpoint by exact coordinates or by radius. The output is a Global Risk-over-time chart and a complete, unpaginated table of every matched event's date, Global Risk score, distance from the query point, EventRootCode, and QuadClass — alongside peak-risk and average-risk summary figures for the checkpoint.

5. Mapping to the Suggested Building Blocks

Set against the five illustrative components in Section 2.3, this iteration of INDRA implements the risk-intelligence layer in full and lays partial groundwork for the digital twin. The remaining components are scoped as future work rather than attempted in a shallow, incomplete form.

Suggested Building Block	Status	Notes
Geopolitical Risk Intelligence Agent	Implemented	GDELT-driven Global Risk scoring, run continuously against any refreshed event export.
Disruption Scenario Modeller	Planned	Historical QuadClass trend analysis at the Strait of Hormuz checkpoint (see Section 6).

Adaptive Procurement Orchestrator	Out of current scope	Not addressed in this iteration.
Strategic Reserve Optimisation Agent	Out of current scope	Not addressed in this iteration.
Supply Chain Digital Twin	Partial	Checkpoint- and route-level risk mapping is a first step toward a full network twin.

Table 3 — Current coverage against the suggested building blocks.

6. Forecasting & Dynamic Rerouting

The core limitation of a point-in-time checkpoint check is the shipment lead time itself: an import decision made today reflects conditions today, but the cargo will not clear the checkpoint for 4–6 weeks. A route cleared at booking can develop an active conflict condition two weeks in, by which point rerouting is costly and reactive rather than planned.

Static screening cannot fully solve this — no model can guarantee 6 weeks of geopolitical foresight from event data alone. Two complementary mitigations are proposed. First, dynamic rerouting: because the engine can be re-run against a refreshed GDELT export at any point in the shipment window, a route can be re-checked and swapped as new events accumulate, rather than being fixed at the original booking decision. Second, short-term forecasting: because three of the five candidate routes converge on the Strait of Hormuz checkpoint, that single location is the highest-value target for a historical trend model. Analysing the recent trajectory of QuadClass distribution and Global Risk at Hormuz — rather than any single event — is the proposed basis for a short-horizon risk forecast, giving the engine an early-warning signal ahead of a checkpoint crossing the 55-point threshold rather than only a real-time reading of it.

Now this is the code to fetch out all the Datasets.

```

def fetch_gdelt_hormuz_data() -> pd.DataFrame:
    """
    Fetches daily event counts related to the Strait of Hormuz
    from the GDELT v2 Event database using Google BigQuery.
    """
    client = bigquery.Client()

    # Optimized SQL Query: Groups by day to keep the payload lightweight
    query = """
        SELECT
            PARSE_DATE('%Y%m%d', CAST(SQLDATE AS STRING)) as ds,
            COUNT(GLOBALEVENTID) as y
        FROM
            `gdelt-bq.gdeltv2.events`
        WHERE
            ActionGeo_FullName LIKE '%Hormuz%'
            OR ActionGeo_FullName LIKE '%Strait of Hormuz%'
        GROUP BY
            ds
        ORDER BY
            ds ASC
    """

    print("Executing BigQuery... (This may take a moment)")
    df = client.query(query).to_dataframe()

    # Drop any null dates that might have bypassed parsing
    df = df.dropna(subset=['ds'])
    return df

```

This model is trained on historical data (`df: pd.DataFrame`) and can forecast future values for up to 4–6 weeks. In the current implementation, the forecast horizon is set to `forecast_days: int=40`, but this value can be modified as needed.

```

def predict_future_events(df: pd.DataFrame, forecast_days: int = 40):
    """
    Trains a Prophet model on the historical GDELT data and
    predicts the event volume for the specified future period.
    """
    # Initialize the Prophet model
    # (yearly_seasonality captures geopolitical cycles, daily is turned off for daily-aggregated data)
    model = Prophet(yearly_seasonality=True, daily_seasonality=False)

    # Fit the model to our historical dataset
    model.fit(df)

    # Create a future dataframe
    future = model.make_future_dataframe(periods=forecast_days)

    # Generate the prediction
    forecast = model.predict(future)

    return model, forecast

```


Now this is the main function to Fetch the Data , then Train the model and Predict then we get all those things we need..

```
if __name__ == "__main__":
    # 1. Fetch the data
    historical_data = fetch_gdelt_hormuz_data()
    print(f"Successfully fetched {len(historical_data)} days of historical data.")

    # 2. Train the model and predict
    print("Training Prophet model and forecasting future events...")
    model, forecast_data = predict_future_events(historical_data, forecast_days=90)

    # 3. Output the results
    print("\nForecast for the next 5 days:")
    # yhat = predicted value, yhat_lower/upper = confidence intervals
    print(forecast_data[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].tail())
```

7. Impact & Feasibility

With close to 90% of national crude supply arriving by import and nearly half of that volume dependent on a single strait, the cost of routing through an unscreened corridor is not abstract — it shows up directly in freight risk premiums, insurance costs, and the delay costs of a shipment caught in a disruption. A checkpoint-level screening layer, even without full forecasting, gives planners a quantified basis to avoid the worst of that exposure before committing a shipment.

On feasibility, both tools run entirely in-browser against a cleaned GDELT export with no server or paid infrastructure required, making the approach inexpensive to pilot and straightforward to hand to a non-technical planning team.

8. Limitations

- No true multi-week forecasting yet — the current system reads present and historical risk, not a projected future state.
- Accuracy is bounded by GDELT's own event coverage and geocoding precision, which is coarser in some regions than others.
- Candidate routes and their checkpoints are currently defined manually rather than derived automatically from shipping/AIS data.
- The 55-point clearance threshold and the scoring weights in Section 3.3 are heuristic starting points, not yet calibrated against real freight, insurance, or delay outcomes.

9. Conclusion

INDRA addresses the risk-intelligence layer of the problem statement directly: a transparent, checkpoint-level Global Risk score derived from GDELT event data, applied consistently across every candidate route into India's energy supply chain. It gives planners a concrete, inspectable answer to "is this checkpoint safe to route through right now" — the foundation the remaining suggested components (scenario modelling, procurement orchestration, reserve optimisation, and the full digital twin) would build on next.

